3 A light helical spring is suspended vertically from a fixed point as shown in Fig. 3.1.

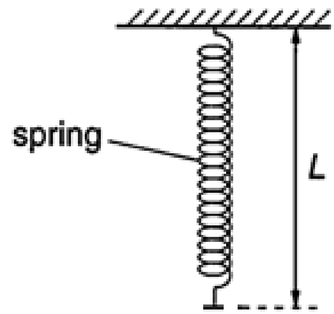


Fig. 3.1

A mass of weight 5.0 N is suspended from the spring of unstretched length 4.0 cm and then released from rest. The mass oscillates vertically.

The variation with resultant force *F* on the mass when *L* is between 4.0 cm and 8.0 cm is shown in Fig. 3.2 below.

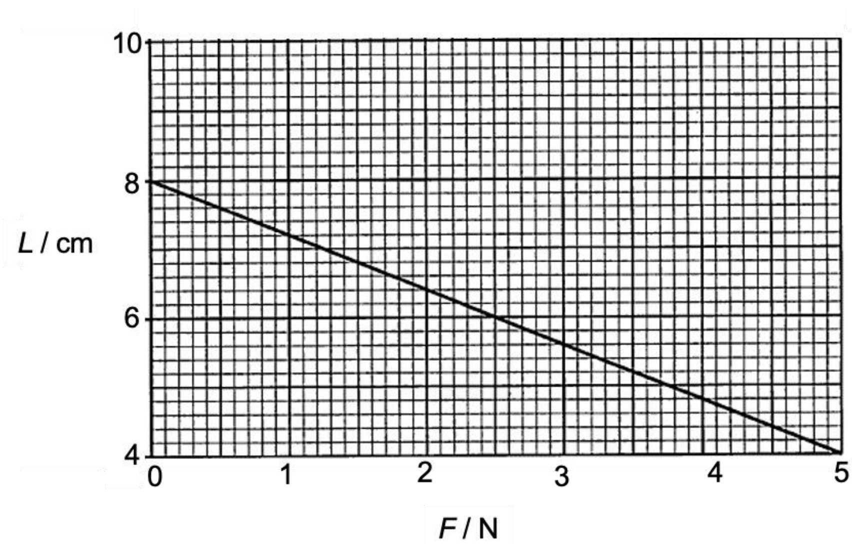


Fig. 3.2

(a) Explain why, as shown in Fig. 3.2, the resultant force on the mass increases as the length of the spring decreases from *L* = 8.0 cm to *L* = 4.0 cm.

....

.... [2]

Fig. 3.2 shows the part of the motion when the mass is **at and above** the equilibrium position.
Also, for the mass to **oscillate,** the upward spring force F_s must be **lesser than** the weight W when the mass is above the equilibrium position.

As spring length *L* decreases, the extension *x* of the spring decreases, hence spring force F_s decreases. 1

Since weight *W* remains constant throughout the motion, the **magnitude of the resultant force *F* acting downwards increases (since F = W – F_s)**. 1

(Hence Fig. 3.2 has a negative gradient.)

(b) Calculate the force constant of the spring.

force constant = ……………………… N m^{-1} [3]

At equilibrium position (where resultant force F = 0),

From Fig. 3.2 → L = 8.0 cm

Since the unstretched spring length = 4.0 cm,

<u>Extension x</u> of the spring at the <u>equilibrium</u> position = 8.0 – 4.0 cm = 4.0 cm **1**

Consider equilibrium of forces:

$$\boldsymbol{F_s = W}$$

$$\boldsymbol{kx = W}$$ **1**

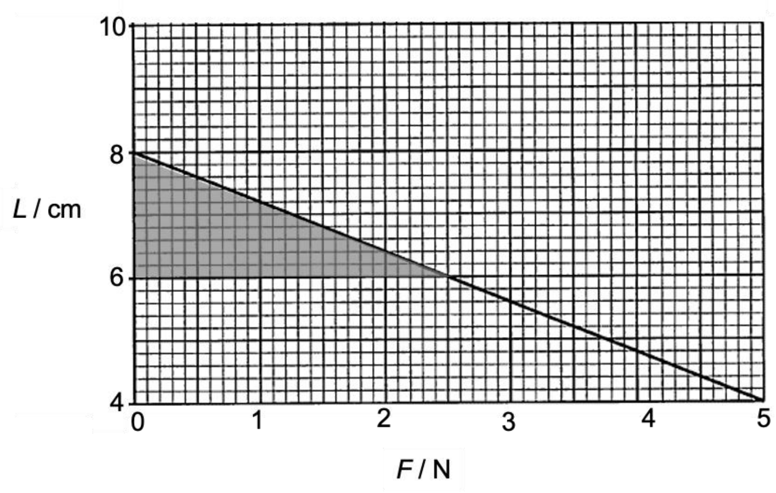
$$\boldsymbol{k = \frac{W}{x}}$$

$$\boldsymbol{= \frac{5.0}{4.0 \times 10^{-2}}}$$ **1**

$$\boldsymbol{= 125\ N\ m^{-1}}$$

(c) On Fig. 3.2, shade clearly the area of the graph that represents net work done on the mass when the mass has travelled from L = 8.0 cm to L = 6.0 cm. [1]

Solution: **1**



For info:

$$WD_{net} = Area\ bounded\ by\ the\ graph\ and\ the\ L - axis$$

Net work done on the mass = $\int F\ ds$
where s is the displacement moved/change in displacement, hence corresponds to the **change in** L.

(d) Describe the energy changes in the spring-mass system when the mass moves from L = 8.0 cm to L = 6.0 cm.

……………………………………………………………………………………………… [2]

The **kinetic energy of the mass decreases and the elastic potential energy of the spring also decreases**, **1**
these energies convert into the increase in the gravitational potential energy of the mass. **1**

4	**(a)**	A revolving aluminium disc has small magnets equally spaced around its rim as shown in